

Practice Quiz 05C — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 05

Checking the assumption in a contradiction proof.

A student claims that, for all real numbers x, y ,

$$x + y \neq 0 \longrightarrow (x \neq 0 \wedge y \neq 0).$$

The proposed proof says: *Suppose $x + y \neq 0$. For a contradiction, assume $x = 0$ and $y = 0$. Then $x + y = 0$, a contradiction. Therefore $x \neq 0$ and $y \neq 0$.*

1. Identify the incorrect negation and write the correct negation of the proposed conclusion.

Solution The negation of $x \neq 0 \wedge y \neq 0$ is $x = 0 \vee y = 0$, not $x = 0 \wedge y = 0$. The proposed proof rules out only the possibility that both numbers are zero. It does not rule out one being zero.

2. Give real values of x, y that disprove the student's claim.

Solution Take $x = 1$ and $y = 0$. Then $x + y = 1 \neq 0$, but $x \neq 0 \wedge y \neq 0$ is false because $y = 0$. Thus the implication is false. ■